MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information معلومات المادة الدراسية					
Module Title	**Data Structures**		**Module Delivery**		
Module Type	**Core**		☒ Theory ☒ Lecture ☒ Lab ☐ Tutorial ☒ Practical ☐ Seminar		
Module Code	**COMP206**				
ECTS Credits	**5**				
SWL (hr/sem)	**125**				
Module Level	2		**Semester of Delivery**		3
Administering Department	CS		**College**	CS and Mathematics	
Module Leader	Names		**e-mail**	Names	
Module Leader's Acad. Title	Assistant Professor		**Module Leader's Qualification**		Ph.D.
Module Tutor	Names		**e-mail**	Names	
Peer Reviewer Name	Name		**e-mail**	E-mail	
Scientific Committee Approval Date	01/06/2023		**Version Number**	1.0	

Relation with other Modules العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	None	**Semester**	
Co-requisites module	Numerical Algorithms	**Semester**	4
Co-requisites module	Algorithm Design and Analysis	**Semester**	5
Co-requisites module	Computer Networks & Communications	**Semester**	6
Co-requisites module	Artificial Intelligence	**Semester**	7

Module Aims, Learning Outcomes and Indicative Contents

<table>
<tr><th colspan="2">أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية</th></tr>
<tr><td>Module Objectives
أهداف المادة الدراسية</td><td>The choice of appropriate data structures is key in the development of efficient algorithms. In fact, it is virtually impossible to create efficient algorithms without a good understanding of a number of fundamental data structures. This module aims
1. To learn how the choice of data structures and algorithm design methods impacts the performance of programs.
2. To learn object-oriented programming principles.
3. To study specific data structures such as linked lists, stacks, queues, hash tables, binary trees, binary search trees, and graphs.
4. To study specific algorithm design method: divide and conquer
5. To gain experience writing programs in Java.</td></tr>
<tr><td>Module Learning Outcomes

مخرجات التعلم للمادة الدراسية</td><td>• Knowledge and understanding
– understand basic ideas about algorithms and time complexity.
– understand the concepts of time and space complexity, worst case, average case and best case complexities and the big-O notation
– know and understand the principles searching and sorting algorithms
– Understand the principles of stacks, queues, linked list, trees, graphs, hashing
• Cognitive skills (thinking and analysis).
– be able to develop efficient algorithms for simple computational tasks
– understand the concepts of time and space complexity, worst case, average case and best case complexities and the big-O notation
– be able to compute complexity measures of algorithms, including recursive algorithms using recurrence relations
– be able to construct and use the data structures such as stacks, queues, linked lists, trees, and hash tables.
– know and understand the principles searching and sorting algorithms
• Communication skills (personal and academic).
• Practical and subject specific skills (Transferable Skills).
– be able to construct and use the data structures such as stacks, queues, linked lists, trees, and hash tables.
– know and understand the principles searching and sorting algorithms
– be able to write advanced programs in Java.</td></tr>
<tr><td>Indicative Contents
المحتويات الإرشادية</td><td>This course tends to give the student the concepts and applications of the data structu including Stacks, Queues, Linked List, Trees, Graphs, and the main Sorting techniques indexing. Object Oriented Programming Language (Java) is used to implement the requ data structures.</td></tr>
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<tr><th colspan="2">Learning and Teaching Strategies
استراتيجيات التعلم والتعليم</th></tr>
<tr><td>Strategies</td><td>The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, practical, lab hours and by considering types of some small projects that are interesting to the students.</td></tr>
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Student Workload (SWL) الحمل الدراسي للطالب محسوب لـ ١٥ أسبوعا			
Structured SWL (h/sem) الحمل الدراسي المنتظم للطالب خلال الفصل	80	**Structured SWL (h/w)** الحمل الدراسي المنتظم للطالب أسبوعيا	7
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطالب خلال الفصل	45	**Unstructured SWL (h/w)** الحمل الدراسي غير المنتظم للطالب أسبوعيا	6
Total SWL (h/sem) الحمل الدراسي الكلي للطالب خلال الفصل	**125**		

Module Evaluation تقييم المادة الدراسية					
		Time/Number	**Weight (Marks)**	**Week Due**	**Relevant Learning Outcome**
Formative assessment	**Quizzes**	2	10% (10)	5 and 10	All
	Assignments	2	10% (10)	4 and 12	All
	Projects / Lab.	1	10% (10)	Continuous	All
	Report	1	10% (10)	13	All
Summative assessment	**Midterm Exam**	2hr	10% (10)	7	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus) المنهاج الاسبوعي النظري	
	Material Covered
Week 1	Introduction to data structures: data structures and algorithms
Week 2+3	Algorithm complexity.
Week 4	Stacks ADT and Stack implementation
Week 5	Stack applications
Week 6	List ADT : static implementation and dynamic implementation, single linked list
Week 7	**First Exam** Lists : doubly linked list and circular linked list.
Week 8	Stacks : Static implementation and dynamic implementation
Week 9	Queues : Static implementation and dynamic implementation, circular queue

Week 10	Trees : Binary search tree, binary expression tree, and heap tree
Week 11	Graph ADT
Week 12	Second Exam Sorting : Bubble sort, selection sort, insertion sort, Shell sort
Week 13	Sorting : Quick sort, Heap sort
Week 14	Searching: Sequential search, Binary Search
Week 15	Hashing : hash function
Week 16	Final Exam

Delivery Plan (Weekly Lab. Syllabus)
المنهاج الاسبوعي للمختبر

	Material Covered
Week 1	Lab 1: The phases of software development
Week 2	Lab 2: Write java programs and calculate time complexity
Week 3	Lab 3: Design and implement Stack using Java codes
Week 4	Lab 4: Applications on stack
Week 5	Lab 5: Create different types of single linked list using java program
Week 6	Lab 6: Write methods to create double linked list and then write program to implement this List ADT
Week 7	First Exam
Week 8	Lap 7: Design and implement queue and circular queue using Java codes
Week 9	Lab 8: Create different types of Trees using Java programs
Week 10	Lab 9: Design graph using Java code
Week 11	Lab 10: Apply all types of sorting algorithms using methods in Java programs
Week 12	Second Exam
Week 13	Lab 11: Write methods in java programs to implement all searching algorithms
Week 14	Lab 12: Discussion Assignments
Week15	Final Exam

Learning and Teaching Resources
مصادر التعلم والتدريس

	Text	**Available in the Library?**
Required Texts	• Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition. • Data Structures & Algorithms In JAVA, 2nd Edition	Yes
Websites	• Data Structures in Java | Beginners Guide |2023 - Great Learning (mygreatlearning.com) • DSA using Java Tutorial (tutorialspoint.com)	

Grading Scheme مخطط الدرجات				
Group	**Grade**	التقدير	**Marks %**	**Definition**
Success Group (50 - 100)	**A -** Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C - Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings
	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	**FX –** Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required

Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.